

Mohammad Niroomand

Department of Computer Engineering,
K. N. Toosi University of Technology, Tehran, Iran

mohammadniroomand80@gmail.com
Website: mohammadniroomand.com
GitHub: github.com/niroomandmd
LinkedIn: [mohammadni](https://www.linkedin.com/in/mohammadni)
Phone: +98 912 096 3018

RESEARCH INTERESTS

- Wireless and Mobile Networks
- 5G & 6G Networks
- Multimedia Networks
- Network Security
- Reinforcement Learning

EDUCATION

K. N. Toosi University of Technology

Sep 2024 – Current

M.Sc. in Computer Engineering, Minor in Computer Networks

Tehran, Iran

Courses: Advanced Computer Networks, Wireless Networks, Multimedia Networks, Performance Evaluation of Computer Systems, Network Management, Advanced Network Security, Cloud Computing, Distributed Systems

Seminar Title: The Role of Non-Terrestrial Networks (NTN) in the Expansion of 5G and Beyond.

Master's Thesis (Supervisor: [Prof. Kaebbeh Yaeghoobi](#)):

- Thesis Title: Performance optimization between terrestrial 5G and LEO satellite networks using reinforcement learning.

University of Guilan

Sep 2019 – Jul 2024

B.Sc. in Computer Engineering

Rasht, Iran

Notable Courses: Computer Networks (4/4), Embedded and Real-Time Systems (4/4), Special Topics 1 (Internet of Things) (4/4), Internet Engineering (4/4), Computer Architecture (4/4), Algorithms Design (4/4), Signal and Systems (4/4)

Bachelor's Project (Supervisor: [Prof. Mohammad Salehi](#)):

- Project summary: Development of a responsive web based weather client application, and leveraging AccuWeather API for real-time Weather data retrieval, forecast visualization and user centric interactive features, together with their design.
- GPA: 4/4

Tadbir High School

Sep 2018 – Sep 2019

High School Diploma in Mathematics & Physics

Qazvin, Iran

- Major GPA: 4/4 (19.10/20)

HONORS AND AWARDS

- Accepted into the Master's program at K. N. Toosi University of Technology in Tehran, **one of the top three institutes of technology** for Computer Networks and Telecommunication Systems in Iran.
- **Ranked in the top 3 %** among more than 16,000 participants in the Iranian National University Entrance Exam (Konkour) for M.Sc. of Computer Engineering, February 2024.
- **Ranked in the top 4 %** among more than 150,000 participants in the Iranian National University Entrance Exam (Konkour) for B.Sc. of Computer Engineering, July 2019.
- Received competitive job offers as **Network Specialist at the Central Bank of Iran** and **Infrastructure Engineer at MTN Irancell**.

- **National LEGO Robotics Team Member** & Qualifier for the 2015 World Robot Olympiad (WRO).
- **Full tuition waiver scholarship** - University of Guilan, 2019-2024

IN-PREPARATION PUBLICATIONS

1. Project Title: **“Federated Trust and Resilience Framework for Sustainable Internet of Vehicles”**
Paper Conclusion: In this study, an approach based on a federated trust and resilience framework for sustainable IoV in 5G/6G was developed. Through the combination of blockchain technology, federated learning, and cross-layer security approaches, this work helps to address many challenges associated with cybersecurity, scalability, privacy protection, and energy efficiency in IoT-based transportation ecosystems. With the implementation of the suggested architecture, it is possible to provide decentralized, intelligent threat mitigation and achieve sustainable, resilient vehicular communication networks. In addition, this work is in line with the strategic objectives of the Vice Presidency of Iran concerning the development of national digital infrastructure, international cooperation in science, and the improvement of intelligent transportation systems based on AI.

ACADEMIC PROJECTS

- **Router Queue Performance Analysis** [\[Link\]](#)
 Simulated and analytically evaluated router queuing systems to analyze packet delay and loss using queuing theory.
- **Advanced Network Security Lab** [\[Link\]](#)
 The five project components were implemented and simulated, including: OSINT & phishing simulation, Nmap scanning, Snort IDS analysis, DoS testing, reverse shell with traffic obfuscation, and HTTPS/TLS certificate & cipher-suite assessment.
- **Hardware Software Co-Design** [\[Link\]](#)
 Design and Implementation of a 6-bit Processor with Programming Capability.
- **Embedded and Real-Time Systems** [\[Link\]](#)
 Design and simulate an air conditioning system using Proteus. Measure temperature and humidity from three different points using sensors and control a heater, a cooler, and a humidifier accordingly.
- **Microprocessors and Assembly Language** [\[Link\]](#)
 Design a traffic light control system at an intersection using microprocessors and assembly language. Simulate the lights in Proteus with LEDs and implement a 7-segment countdown timer.
- **Sentiment Analysis** [\[Link\]](#)
 Build both traditional machine learning models (Naïve Bayes/Logistic Regression) and a TensorFlow neural network for sentiment analysis on the Amazon reviews dataset.

TEACHING EXPERIENCE

- **Teaching Assistant** at K. N. Toosi University of Technology
 - Network Management (Spring 2026) Sep 2025 - Currently
Supervisor: Prof. Kaebbeh Yaeghoobi
 - Computer Networks I (Spring 2026) Supervisor: Prof. Kaebbeh Yaeghoobi
 - Digital Electronics (Spring 2026) Supervisor: Prof. Zeinab Seifoori
 - Technical English (Spring 2026) Supervisor: Prof. Kaebbeh Yaeghoobi
 - Advanced Computer Networks (Fall 2025) Supervisor: Prof. Kaebbeh Yaeghoobi
- **Teaching Assistant** at University of Guilan
 - Embedded and Real-Time Systems (Spring 2024) Sep 2019 - Jul 2024
Supervisor: Prof. Mohammad Salehi
 - Algorithm Design (Spring 2023) Supervisor: Prof. Aida Khozaei
 - Advanced Programming (Spring 2022) Supervisor: Prof. Aida Khozaei

COURSES

- **Sample-based Learning Methods** | [Verification](#) May 2026
University of Alberta
- **Fundamentals of Reinforcement Learning** | [Verification](#) Dec 2025
University of Alberta
- **CISSP Certified Information Systems Security Professional** | [Verification](#) Jul 2025
Packt
- **Supervised Machine Learning: Regression and Classification** | [Verification](#) Dec 2024
DeepLearning.AI
- **CCNA 200-301** | [Verification](#) Nov 2024
Arjang Institute of Higher Education
- **CEH (Certified Ethical Hacker)** | [Verification](#) Apr 2024
Maktabkhooneh
- **LEGO Education and Advanced Robotics Training** | [Verification](#) 2014-2015
LEGO Education Institute

SKILLS

- **Programming Languages:** Python, Java, C/C++, VHDL/Verilog, Solidity
- **Developer Tools:** EVE-NG, GNS3, Cisco Packet Tracer, Paessler PRTG, Wireshark, Proteus, VS Code, Linux, PyCharm, IntelliJ, Git, Github, Slack, $\text{\LaTeX}$
- **Web Development:** React, Bootstrap, HTML/CSS, Tailwind CSS, JavaScript, MySQL
- **English:** IELTS: To be taken soon

PERSONAL INTERESTS

- **Music:** Professional piano player with two years of teaching experience.
- **Literature & Arts:** Interested in poetry and literature, with experience in film, theater, and book criticism. Furthermore, I organized an open studio in Cité Internationale des Arts in Paris (2015)
- **Social & Environmental Engagement:** Active collaboration with charitable and environmental institutions.